

Data Transformation and data modelling

Project Objective: The objective of this assignment is to import data, cleaning and relationship modelling using Power BI and power query editor.

Dataset Description: This is the E-commerce sales dataset in it we have three datasets namely, List of orders, Order details, Sales target.

Column Description:

List of orders:

Order ID	Displays unique ID for each order
Order Date	Display date in format
Customer Name	Displays name of the customer
State	State of the customer
City	City of the customer

Order Details:

Order ID	Order ID of the customer
Amount	Amount of the order
Profit	Profit of the order
Quantity	How much quantity in numbers
Category	Category of the product

Sub category	Sub category of the product
--------------	-----------------------------

Sales Target:

Month of Order Date	Displays month with year
Category	Category of the product
Target	Target amount for the product

Data Transformation and Data Modelling:

1.Data Import:

List of Order,Order Details,List of Orders Table are loaded into POWER BI which are csv files and they are transformed and converted into table using **Transform Data in Power Query Editor.**

2.Row Limiting and Data types:

a)Restrict List of orders table to first 500 rows:The table rows are restricted to 500 rows using remove columns option.

b)Order date column to date data type:Data type is changed using Transform option in power query editor.

c)Change amount and target columns to fixed decimal numbers:These columns are changed to fixed decimal numbers option using Transform option.

3.Text Formatting:

Format customer name column to Proper case:Customer name is formatted using **Proper option to capitalize word.**

4.Column Creation:

City and state column is merged and location column is created using Custom column option.

	ABC 123 Location
	Gujarat , Ahmedabad
	Maharashtra , Pune
	Madhya Pradesh , Bhopal
	Rajasthan , Jaipur
	West Bengal , Kolkata
	Karnataka , Bangalore
	Jammu and Kashmir , Kashmir
	Tamil Nadu , Chennai
	Uttar Pradesh , Lucknow
	Bihar , Patna
uram	Kerala , Thiruvananthapuram
	Punjab , Chandigarh
	Haryana , Chandigarh
	Himachal Pradesh , Simla
	Sikkim , Gangtok
	Goa , Goa
	Nagaland , Kohima
	Andhra Pradesh , Hyderabad
	Gujarat , Ahmedabad
	Maharashtra , Pune
	Madhya Pradesh , Bhopal

Create custom column named Profit margin=Profit/Amount and convert it into percentage.

	Margin Profit
	37.79%
	-4.20%
	-86.01%
	3.18%
	0.00%
	-20.00%
	0.00%
	-36.88%
	0.00%
	25.50%
	0.00%
	-58.39%
	-54.96%
	-37.50%
	-1.13%
	7.85%
	17.74%
	-38.52%
	-40.00%
	-14.29%
	-47.27%
	-31.54%
	-49.76%
	-6.45%
	-43.95%
	8.40%

5.Conditional Column:

Conditional column named Profit Status is added using condition:

Profit<0 Loss

Profit=0 Break-even

Profit>0 Profit

ABC 123 Profit margin

Loss
Loss
Loss
Loss
Loss
Loss
profit
profit
Loss
Loss
Loss
Loss
profit
profit
profit

6.Merging data:

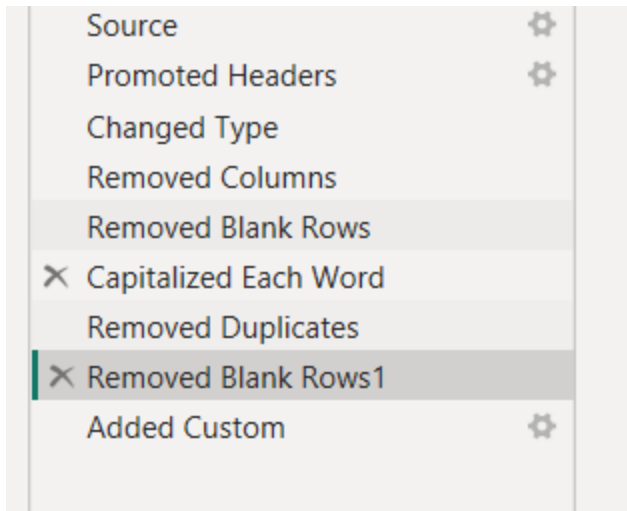
List of orders and order details table is merged using Order ID and it is named as Order Data using Merge Queries option.

Table.NestedJoin(#"List of orders", {"Order ID"}, #"Order Details", {"Order ID"}, "Order Details", JoinKind.LeftOuter)

Order ID	CustomerName	State	City	Location	Order
B-25601	Bharat	Gujarat	Ahmedabad	Gujarat , Ahmedabad	Table
B-25602	Pearl	Maharashtra	Pune	Maharashtra , Pune	Table
B-25603	Jahan	Madhya Pradesh	Bhopal	Madhya Pradesh , Bhopal	Table
B-25604	Divsha	Rajasthan	Jaipur	Rajasthan , Jaipur	Table
B-25605	Kasheen	West Bengal	Kolkata	West Bengal , Kolkata	Table
B-25606	Hazel	Karnataka	Bangalore	Karnataka , Bangalore	Table
B-25607	Sonakshi	Jammu and Kashmir	Kashmir	Jammu and Kashmir , Kashmir	Table
B-25608	Aarushi	Tamil Nadu	Chennai	Tamil Nadu , Chennai	Table
B-25609	Jitesh	Uttar Pradesh	Lucknow	Uttar Pradesh , Lucknow	Table
B-25610	Vogesh	Bihar	Patna	Bihar , Patna	Table
B-25611	Anita	Kerala	Thiruvananthapuram	Kerala , Thiruvananthapuram	Table
B-25612	Shrichand	Punjab	Chandigarh	Punjab , Chandigarh	Table
B-25613	Mukesh	Haryana	Chandigarh	Haryana , Chandigarh	Table
B-25614	Vandana	Himachal Pradesh	Simla	Himachal Pradesh , Simla	Table
B-25615	Bhavna	Sikkim	Gangtok	Sikkim , Gangtok	Table
B-25616	Kanak	Goa	Goa	Goa , Goa	Table
B-25617	Sagar	Nagaland	Kohima	Nagaland , Kohima	Table
B-25618	Manju	Andhra Pradesh	Hyderabad	Andhra Pradesh , Hyderabad	Table
B-25619	Ramesh	Gujarat	Ahmedabad	Gujarat , Ahmedabad	Table
B-25620	Sarita	Maharashtra	Pune	Maharashtra , Pune	Table
B-25621	Deepak	Madhya Pradesh	Bhopal	Madhya Pradesh , Bhopal	Table
B-25622	Monisha	Rajasthan	Jaipur	Rajasthan , Jaipur	Table
B-25623	Atharv	West Bengal	Kolkata	West Bengal , Kolkata	Table
B-25624	Vini	Karnataka	Bangalore	Karnataka , Bangalore	Table
B-25625	Pinky	Jammu and Kashmir	Kashmir	Jammu and Kashmir , Kashmir	Table
B-25626	Bhishm	Maharashtra	Mumbai	Maharashtra , Mumbai	Table
B-25627	Hitika	Madhya Pradesh	Indore	Madhya Pradesh , Indore	Table
B-25628	Pooja	Bihar	Patna	Bihar , Patna	Table
B-25629	Hemant	Kerala	Thiruvananthapuram	Kerala , Thiruvananthapuram	Table
B-25630	Sahil	Punjab	Chandigarh	Punjab , Chandigarh	Table
B-25631	Ritu	Haryana	Chandigarh	Haryana , Chandigarh	Table
B-25632	Manish	Himachal Pradesh	Simla	Himachal Pradesh , Simla	Table

7.Handling missing data and duplicate data:

There is no missing values in dataset but duplicate values are removed using remove duplicates. empty rows are removed which has null value.



8.Sorting and filtering data:

Records in order data table are sorted using descending order [option.If](#) we right click the specific column we can sort either ascending or descending order.

We can apply filters using right click for specific state called TamilNadu.

Order ID	CustomerName	State	City
B-26081	Aarushi	Tamil Nadu	Chennai
B-26018	Aarushi	Tamil Nadu	Chennai
B-26008	Kalyani	Tamil Nadu	Chennai
B-25860	Akshay	Tamil Nadu	Chennai
B-25788	Dinesh	Tamil Nadu	Chennai
B-25716	Surabhi	Tamil Nadu	Chennai
B-25698	Amisha	Tamil Nadu	Chennai
B-25608	Aarushi	Tamil Nadu	Chennai

9.Grouping and Aggregating data:

In order details table,we can create Duplicates by right clicking the table.

a)Count of Order Id:

formula used:=COUNT(order Id)

Order count=1500.

b)Average Profit by category:Average profit is calculated categorywise.

Data	
Average Profit	Category
34.07	Electronics
11.76	Clothing
9.46	Furniture
15.97	

c) Total amount by sub-category:Sum of the amount is calculated for different sub-categories.

Sum	Sub-Category
₹ 58,252	Printers
₹ 56,861	Bookcases
₹ 53,511	Saree
₹ 46,119	Phones
₹ 39,168	Electronic Games
₹ 34,222	Chairs
₹ 30,039	Trousers
₹ 22,614	Tables
₹ 21,728	Accessories
₹ 18,546	Stole
₹ 4,31,502	

Sales target table is duplicated and total target by month of order date is calculated.

Month of Order Date	Category	Target
01 April 2018	Furniture	₹ 10,400
01 May 2018	Furniture	₹ 10,500
01 June 2018	Furniture	₹ 10,600
01 July 2018	Furniture	₹ 10,800
01 August 2018	Furniture	₹ 10,900
01 September 2018	Furniture	₹ 11,000
01 October 2018	Furniture	₹ 11,100
01 November 2018	Furniture	₹ 11,300
01 December 2018	Furniture	₹ 11,400
01 January 2019	Furniture	₹ 11,500
01 February 2019	Furniture	₹ 11,600
01 March 2019	Furniture	₹ 11,800

10.Data Modeling:

a)Relationship between List of order and order details using **Order Id**.

← Edit relationship

×

Select tables and columns that are related.

From table

Order Details (2)

Category	Order ID	Profit	Quantity	Sub-Category
Clothing	B-25602	212	3	Saree
Clothing	B-25602	-5	8	Saree
Clothing	B-25603	-166	3	Saree

To table

List of Orders

City	CustomerName	Order ID	State
Indore	Hitika	B-25627	Madhya Prad...
Indore	Ashmi	B-25637	Madhya Prad...
Indore	Yaanvi	B-25645	Madhya Prad...

Cardinality

Many to one (*:1)

Cross-filter direction

Single

b)Relationship between order details and sales target using **Category**.

← **Edit relationship**

Select tables and columns that are related.

From table

Order Details

Category	Category type	Order ID	Profit	Quantity
Clothing	Clothing,Saree	B-25602	212	3
Clothing	Clothing,Saree	B-25602	-5	8
Clothing	Clothing,Saree	B-25603	-166	3

To table

Sales target

Category	Month of Ord...
Furniture	01 April 2018
Furniture	01 May 2018
Furniture	01 June 2018

2.Using Manage relationship to ensure relationship are active and defined correctly.

Manage relationships

+ New relationship Autodetect Edit Delete Filter				
☐	From: table (column) ↑	Relationship ... ↑	To: table (column)	Status
☐	Order Details (2) (Order ID)	* ← 1	List of Orders (Order ID)	Active ...
☐	Order Details (Category)	* ↔ *	Sales target (Category)	Active ...

Conclusion:

For the given E-commerce sales dataset data is imported,transformed and cleaned.Relationship modelling is established for the given three tables to ensure accuracy.

